

Hybrid Predictive–Optimization Modeling of LoRa-Based Emergency Communication Systems for Landslide Mitigation

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1. Background

Landslides are one of the natural disasters with significant global impact. According to studies by Froude and Petley, during the period 2004–2016 there were 4,862 fatal landslide events worldwide, resulting in 56,997 deaths [1][2]. The impacts are not limited to casualties but also include damage to vital infrastructure such as roads, bridges, electricity, and communication systems. To address this issue, LoRa (Long Range) is considered a potential alternative solution. LoRa enables long-distance communication with low power consumption and can continue to function even when cellular infrastructure is damaged, making it possible to transmit data and location information in real time under challenging field conditions.

LoRa technology has been proven to hold great potential for long-range communication with low power consumption [3], although challenges remain in terms of scalability in large-scale networks [4]. One of the key challenges in LoRa implementation is the increased risk of packet collisions as the number of connected devices grows, which reduces communication reliability. Moreover, physical environmental factors such as buildings, vegetation, topography, and weather conditions significantly affect signal propagation, often leading to inconsistent LoRa performance across different locations [3].

Therefore, this research is motivated by the critical need to overcome the environmental limitations of LoRa for emergency response. We propose a novel solution such as an intelligent system that uses predictive modeling and optimization algorithms to automatically determine the best locations for communication beacons and the most reliable paths for data to travel. This ensures a strong and dependable network can be established automatically, even in the challenging terrain of a landslide disaster zone, ultimately saving crucial time and lives.

2. Objectives With Hypothesis

This research aims to develop a hybrid predictive–optimization model to enhance the efficiency and reliability of LoRa-based emergency communication systems in landslide disaster scenarios. The model is designed to automatically recommend strategic beacon placement and the most optimal multi-hop communication routes, while considering the complex and non-linear

topographic characteristics of the terrain. By employing machine learning algorithms (such as Random Forest) trained with empirical field data (RSSI, PDR, latency) as well as spatial data (elevation, land cover), the model predicts communication link quality in unmeasured areas. These predictions then serve as input for a graph-based pathfinding algorithm (such as A*) to identify routes with the strongest signal strength and lowest latency.

The advantages of our proposed will be quantitatively evaluated using key metrics, including improvements in Packet Delivery Ratio (PDR), reductions in latency, expanded coverage area, and cost efficiency through a reduced number of required beacons. Accordingly, this research is expected to produce a framework that demonstrates superior performance and greater resilience to environmental disruptions compared to traditional placement approaches that rely solely on intuition or line-of-sight coverage.

3. Proposed Method

This research began with a systematic empirical data acquisition process using ESP32-LoRa devices integrated with GPS to record crucial communication metrics such as RSSI, PDR, and delay. The collected raw data then underwent a preprocessing stage to remove invalid or incomplete entries. Subsequently, exploratory data analysis and visualization were performed using descriptive statistics and techniques such as scatter plots and heatmaps to understand the characteristics and distribution of the data. This analysis was further enhanced with correlation and regression studies to identify and quantify the impact of environmental variables such as distance and physical obstacles on signal quality, including the investigation of signal propagation anomalies.

The next stage involves the development of a hybrid predictive–optimization model. A predictive model, trained using machine learning algorithms (such as Random Forest), is employed to map the relationship between geographical features (elevation, land cover) and signal quality. The predictions from this model then serve as input for an optimization model based on the A* algorithm, which is designed to determine beacon placement and optimal multi-hop routes with the strongest signal and lowest latency. Model performance is validated through comparative simulations against

standard routing paths, followed by field trials for verification. Accuracy is evaluated using error metrics such as RMSE and MAE, while the overall system performance is assessed based on communication metrics including PDR, latency, and the efficiency of beacon usage.

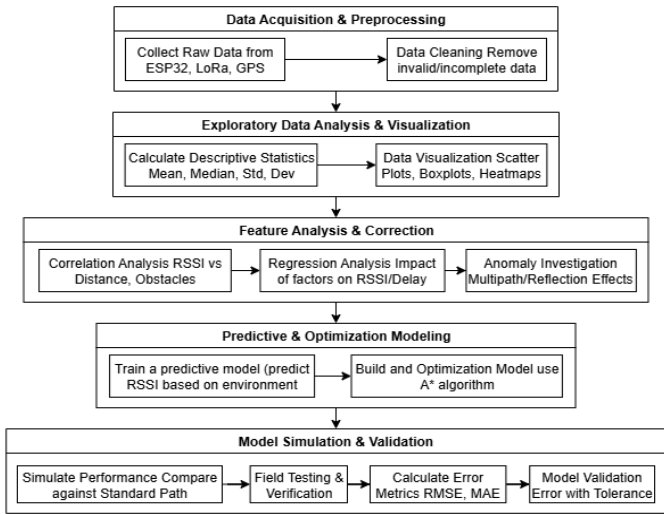


Figure 1. Hybrid Model Overview

4. Expected Result and Discussion

The LoRa-based emergency communication system is expected to expand coverage, reduce latency, and improve beacon usage efficiency compared to Line-of-Sight methods. The hybrid model combines Random Forest for link quality prediction (RSSI, delay) and A* for optimal multi-hop route selection, ensuring stable communication paths in complex terrain conditions. Performance evaluation focuses on Packet Delivery Ratio (PDR), latency, coverage, and cost efficiency. Simulations and field tests are expected to show an increase in PDR, a decrease in latency, and beacon savings of up to 20–30%. Visualizations in the form of RSSI heatmaps and PDR-distance curves will be used to show performance patterns in various environmental conditions.

In addition, our system is designed to be adaptive to data anomalies, extreme weather, multipath, and collisions, enabling it to adjust communication routes in real time. With these advantages, this framework is expected to be more resilient, energy efficient, and replicable for various emergency response scenarios. The resulting framework not only presents technical advances in LoRa network management but also provides practical and applicable solutions to improve preparedness in landslide disaster mitigation. The findings of this study are expected to serve as an important reference in the development of more resilient and efficient emergency communication systems in the future.

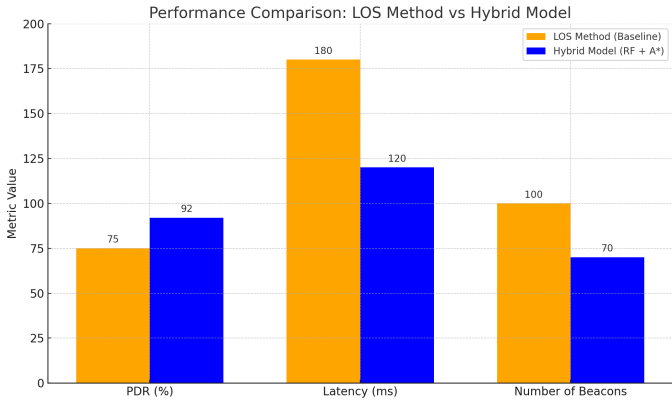


Figure 2. Expected Result and Discussion

5. Timeline

Period	Main Activities	Expected Outcomes
September 2025	Field data collection & initial model design	Dataset & draft model
Late Sept - Early Oct 2025	Data analysis & development of hybrid model	Draft optimization model
Mid October 2025	Simulation & field testing	Trial report
Late October 2025	Model refinement & manuscript writing	Final draft report (Oct 27)

References

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[3] Centenaro, M., Vangelista, L., Zanella, A., & Zorzi, M. (2016). Long-range communications in unlicensed bands: the rising stars in the IoT and smart city scenarios. *IEEE Wireless Communications*, 23(5), 60–67. <https://doi.org/10.1109/MWC.2016.7721743>

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